

PAPER197: FUBi Extended Integral — UV, mm-Wave, Hybrid, and Hierarchical Terms

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Session: 50 — grokshare7514fe.txt Full Audit

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Version: 1.0 **Date:** March 13, 2026 **Session:** 50 — grokshare7514fe.txt Full Audit **Author:** Star-Magic UQFF Research Framework **Source:** grokshare7514fe.txt lines 200–400

$$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{bi}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$$

<!-- ? = 5.0e-4 day⁻¹, [SSq] = 0.57, β_i = 6.1e-1 -->

Abstract

This paper documents the extended form of the FUBi buoyancy integral, incorporating four new terms beyond the standard UQFF formulation: FUV (GALEX/Spitzer UV flare coupling), Fmm (ALMA mm-radio coupling), Fhyb (hybrid polarization-frequency term), and Fhier (hierarchical remnant unification). Numerical coefficients $k_{UV} = k_{mm} = 10^{3^\circ} \text{ N/W}$ and $f_{mm} = 1.05$ are derived from observational calibration. This extended integral enables multi-wavelength coupling of buoyancy forces from UV through millimeter radiation fields.

1. Standard FUBi Integral (Baseline)

The standard buoyancy integral form:

$$F_{U_{Bi}} = -F_0 + (m_e c^2 / r^2) \cdot \text{DPM_momentum} \cdot \cos? + (GM / r^2) \cdot \text{DPM_gravity} + F_{U_{Bi}}$$

2. Extended FUBi Integral

The complete extended form including all observational coupling terms:

$$F_{U_{Bi}} = ?0^{x2} [$$

$$-F_0$$

$$+ (m_e c^2 / r^2) \cdot \text{DPM_mom} \cdot \cos?$$

$$+ (GM / r^2) \cdot \text{DPM_grav}$$

$$+ ?_{vac}, [UA] \cdot \text{DPM_stab}$$

$$+ k_{LENR} \cdot (?_{LENR} / ?0)^2$$

$$+ k_{act} \cdot \cos(?_{act} \cdot t)$$

$$+ k_{DE} \cdot L_X$$

$$+ 2qB0V \cdot \sin? \cdot \text{DPM_res} \cdot P_{pol}$$

$$+ k_{neutron} \cdot s_n$$

$$+ k_{rel} \cdot (E_{cm,astro,enhanced} / E_{cm})^2$$

$$+ k_{UV} \cdot L_{UV} \text{ ? NEW: UV flare coupling}$$

+ k_mm · L_mm · f_mm ? NEW: mm-radio coupling

] dx

3. New Extended Terms

3.1 F_UV — GALEX/Spitzer UV Coupling

F_UV = k_UV · L_UV

Parameters:

k_UV = 10³³ N/W (UV force coupling constant)

L_UV = UV luminosity (W) from GALEX or Spitzer photometry

Physical origin: UV flare irradiation pressure on buoyant plasma shells

3.2 F_mm — ALMA mm-Radio Coupling

F_mm = k_mm · L_mm · f_mm

Parameters:

k_mm = 10³³ N/W (mm-radio force coupling constant)

L_mm = mm-radio luminosity (W) from ALMA observations

f_mm = 1.05 (mm-radio frequency enhancement factor)

Physical origin: millimeter-wave radiation pressure in molecular cloud outflows

3.3 F_hyb — Hybrid Polarization-Frequency Term

F_hyb = P_pol · f_mm · (θ)¹

Parameters:

P_pol = polarization fraction (dimensionless, typically 0.01-0.1)

f_mm = 1.05

θ = base UQFF angular frequency (rad/s)

Physical origin: coupling of polarized mm-emission to buoyancy oscillation frequency

3.4 F_hier — Hierarchical Remnant Unification

F_hier = S · (v_i/c)ⁿ · θ^{-m}

Parameters:

v_i = velocity of remnant component i (m/s)

c = speed of light

n = 2 (hierarchical power index)

m = 1 (frequency suppression exponent)

Physical origin: multi-component remnant velocity hierarchy (e.g., jet + cocoon + lobe)

4. Standard Integral Terms (Reference)

Term	Symbol	Physical Origin
Base restoring force	-F0	Vacuum restoring force
DPM momentum	(me c ² /r ²)·DPMmom·cos?	Dipole-plasma momentum scattering
DPM gravity	(GM/r ²)·DPM_grav	Dipole-plasma gravitational coupling
DPM stability	?vac,[UA]·DPMstab	Vacuum aether stability term
LENR coupling	kLENR·(?LENR/θ) ²	Low-energy nuclear resonance
Activation term	kact·cos(?act·t)	Activation oscillation
Dark energy luminosity	kDE·LX	X-ray dark energy coupling

Term	Symbol	Physical Origin
DPM resonance	$2qB0V \cdot \sin? \cdot DPMres \cdot Ppol$	Magnetic resonance polarization
Neutron cross-section	$kneutron \cdot sn$	Neutron scattering
Relativistic CM	$krel \cdot (Ecm,astro,enh/E_cm)^2$	Relativistic center-of-mass enhancement

5. Numerical Results for Extended Terms

System	Standard FUBi_i	With UV/mm extension
Magnetar SGR1745	$\sim 2.11 \times 10^{2\circ 8} \text{ N}$	+ F_UV from Chandra/GALEX
NGC 3603	$\sim -8.31 \times 10^{211} \text{ N}$	+ F_mm from ALMA CO observations
Pillars of Creation	$\sim 9.79 \times 10^{33} \text{ N}$	+ Fhyb with Ppol ~ 0.05

Note: The extreme magnitudes ($10^{2\circ 8} \text{ N}$, 10^{211} N) reflect vacuum-density-scaled units in the UQFF framework where $?_vac,[UA] \sim 10^{113}$ (dimensionless normalized).

6. Integration with Existing UQFF Terms

The extended integral fits within the Triadic Master System (PAPER196) *as the primary buoyancy channel FUBi*. Specifically:

- F_UV** activates when GALEX UV flux > threshold (flare events)
- F_mm** activates when ALMA observes SFR-driven mm continuum
- F_hyb** operates continuously as polarization coupling
- F_hier** applies to multi-component systems (AGN jets, SNR shells)

7. References

grok_share_7514fe.txt lines 200–400 (first PDF extended integral section)
PAPER_196: Triadic Master Equation System
PAPER_171: Universal Gravity Ug1–Ug4 Decomposition
PAPER_172: FU Complete Unified Field Assembly